PINE RIDGE FIRE: NATIVE WINNER SEEDING TRIALS

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STUDY BACKGROUND: This research was designed to identify how potential "native winner" species perform in post-fire revegetation conditions where cheatgrass is present, relative to the native species and sources that are currently available. We also tested whether the ratio of grasses to forbs impacts the outcome of revegetation efforts in habitat degraded by cheatgrass. For this, we worked with the BLM Grand Junction Field Office (GJFO) following the July 2012 Pine Ridge Fire to establish three study blocks at a site with high percent cheatgrass cover, and monitored percent cover of sown and spontaneous plant species in all blocks in May and September 2013, June 2014, and May 2016. We aimed to identify which of the "native winner" species we sowed are able to germinate, establish, and persist at the site, and compare this with species and sources in the GJFO's Broad Site seed mix. Ultimately, our goal was to help identify seeding sourcing approaches that can improve the long-term outcomes of revegetation efforts in cheatgrass-degraded habitat.

DATE PLOTS ESTABLISHED: Plots were set up the week of October 29th, 2012. Treatments were applied (seed sown) on November 1, 2012.

LOCATION DETAILS: This study was established in the northeast corner of the Pine Ridge Fire site (on the north rim of Horseshoe Canyon) near BLM Frequency Monitoring Point 6712_F (at 39° 15' 59.4" -108° 16' 41.7"). All blocks were established at least 100m from each other in habitat that was reported to have moderate burn severity, and which received the Aerial Seed Broad Site Mix when seed is sown at the site (applied in early 2013). QuickGuard was applied to the site by GJFO prior to our study setup, but hadn't germinated by Nov 1, 2012 except in small gullies created by erosion where soil was slightly moister.

Our three 16 x 8 meter study blocks were set up at the following locations:

1. 39° 15' 56.6" -108° 16' 33.0" elevation 1621m
2. 39° 15' 54.4" -108° 16' 26.9" elevation 1621m
3. 39° 15' 55.7" -108° 16' 54.9" elevation 1644m

STUDY DESIGN: Six treatments were applied to 3 plots each within each study block (See Appendix 1 for species details):

A. <u>BLM Broad Site Mix</u>: ~90% grasses/~10% forbs/~0.3% shrubs. This mix was applied to 3, 2x2m plots in each study block, for a total of 9 study plots. It used all species and sources that will be included in the seed mix applied aerially in early 2013, including 6 grass species (11 sources), 7 forb species (8 sources) and 1 shrub (1 source). Seed was sown at a total rate of ~750PLS/m2, which was the target PLS/m2 rate indicated in the BLM's Burned Area Rehabilitation report (note that the rate that will likely be aerially applied will be closer to 575 based on PLS ratings for purchased seed and pounds of seed purchased).

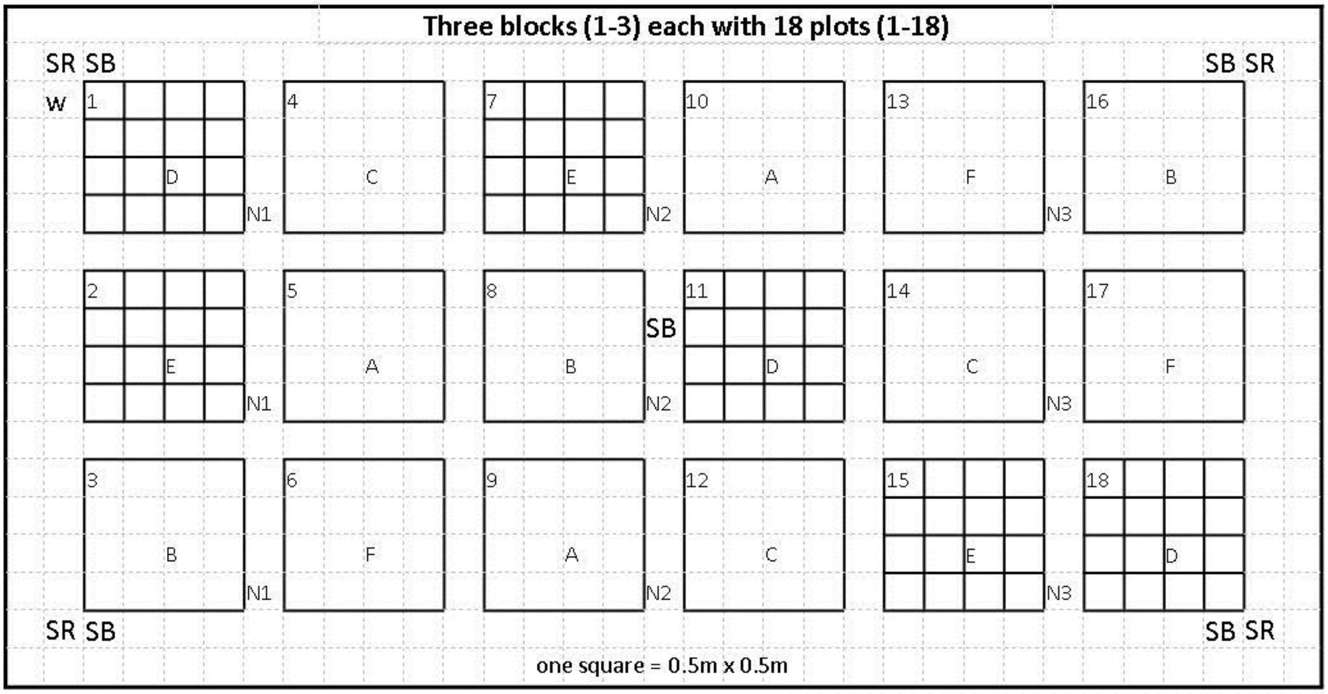
B. <u>BLM/Native Winner Mix</u>: ~90% grasses/~10% forbs/~0.3% shrubs. This mix was applied to 3, 2x2m plots in each study block, for a total of 9 study plots. It included 6 grass species (10 sources; 4 of which were NW and 6 of which were BLM), 14 forbs (14 sources; 11 of which were NW and 3 of which were BLM (from UP)), and 1 shrub (1 source). Seed was sown at a total rate of ~750PLS/m2.

C. <u>BLM/Native Winner Mix High Forb</u>: ~66% grasses/~33% forbs/~0.3% shrubs. This mix was applied to 3, 2x2m plots in each study block. This included 6 grass species (10 sources; 4 of which were NW and 6 of which were BLM), 14 forbs (14 sources; 11 of which were NW and 3 of which were BLM (from UP)), and 1 shrub (1 source). Seed was sown at a total rate of ~750PLS/m2.

D. <u>Forb Monoculture</u>: One of 16 forb species was applied to three 0.5x0.5m sub-plots in each study block, for a total of 9 study sub-plots. This included 6 BLM forbs and 10 NW forbs. Seed was sown at a total rate of ~750PLS/m2.

E. Grass Monoculture: One of 15 grass species and 1 forb species was applied to three 0.5x0.5m sub-plots in each study block, for a total of 9 study sub-plots. This included 11 BLM grasses, 4 NW grasses, and one NW forb. Seed was sown at a total rate of ~750PLS/m2.
F. Control: No seed was sown in this plot.

All treatments were assigned to plots using stratified random design, and seed was mixed prior to sowing. For all plots, soil was lightly disturbed prior to hand-sowing, and seed was lightly smoothed into the plots by hand. Note that it was not possible to cover up these plots, so they all received the aerial application of the Broad Site Seed Mix in early 2013.



Six treatments	
A	BLM MIX
B	BLM/NATIVE WINNER MIX
C	BLM/NATIVE WINNER HIGH FORB MIX
D	GRASS MONOCULTURES
E	FORB MONOCULTURES
F	CONTROL

16 sub-plots / plot

a	b	c	d
e	f	g	h
i	j	k	l
m	n	o	p

Additional details	
w	Wooden stake in top left corner of all plots
SR	Seed rain traps on all four corners (4/block)
N#	Soil samples for nitrogen tests (3 samples/block)
SB	Seed bank samples (1 sample/block)

RESEARCH QUESTIONS ADDRESSED:
1. H1: Incorporation of native winners in a seed mix (BLM/Native Winner Mix, using the same seeding rate and ratio of 9:1 grasses to forbs) delivers BLM objectives better than the BLM Broad Site Mix.
   a. H1a: Cover of native annuals will be greater in BLM/NW plots than the BLM mix (short-term)
   b. H1b: Cover of native perennials will be greater in BLM/NW plots than the BLM mix (long-term)
   c. H1c: Cover of undesirable non-native species (e.g., cheatgrass) will be lower in BLM/NW plots than in the broad site seed mix plots (short- and long-term)
2. H2: Incorporation of native winners and a higher proportion of forbs in a seed mix (BLM/Native Winner High Forb, using the same seeding rate and a ratio of 2:1 grasses to forbs) delivers BLM objectives better than the BLM Broad Site Mix or BLM/Native Winner Mix.
   a. H2a: Cover of native annuals will be greater in BLM/NWHighForb plots than the BLM of BLM/NW plots (short-term)
   b. H2b: Cover of native perennials will be greater in BLM/NWHighForb plots than the BLM of BLM/NW plots (long-term)
   c. H2c: Cover of undesirable non-native species (e.g., cheatgrass) will be lower in BLM/NWHighForb plots than in the broad site seed mix plots (short- and long-term)

3. H3: Diverse species mixes are better at delivering BLM objectives than monocultures (using the same seeding rate) over the long-term (3+ years).
    a. H3a: Percent cover of undesirable non-native species (cheatgrass) will be lower in BLM/NW High Forb mix, BLM/NW, and BLM Broad Site plots than in monoculture plots

DATA COLLECTION AND RESULTS

1. Plot establishment (Oct 30, 2012):
    a. VEGETATION SURVEY: categorize percent cover of bare ground, fine debris, coarse debris, and plants by species for all sub-plots in all 3 study blocks. We also conducted a random walk around the study plots and any pockets of unburned vegetation nearby to identify which species may contribute to seed rain in future years.
    b. SEED RAIN: traps established in all 4 corners of each block to determine whether application of aerially seeded species matches intended rate. For this, flower pots were put in holes so they sit nearly flush with the soil surface. Nylon stockings were put inside the flower pot to trap seeds for identification in May 2013.
    c. SEED BANK: 5 soil cores (6cm dia x 3cm deep) were collected from each corner and the center of each study block and put into a single plastic bag (for each block). The same sampling approach was used to collect soil from 3 nearby plots that did not burn to determine fire impacts on diversity and abundance of seeds in the soil.
2. Seed rain (May 2013):
    a. Traps were checked but were filled with soil sometime between Nov 1 2012 and May 17, 2013. It was therefore not possible to use them to determine whether aerial application had occurred at the intended rate at the site, and they were removed.
3. Seed germination trials (February - June 2013):
    a. We carried out seed germination trials for all species included in these study plots, subjecting them to six different treatments in incubators at Chicago Botanic Garden that simulate different seeding times. Data has been entered into a database, and made available through the Colorado Plateau Native Plant Program Database on the Conservation Registry (http://cpnpp.conservationregistry.org/projects/247716) and published on the Native Plant Propagation database (http://npn.rngr.net/propagation/protocols).
4. Seed bank study (February 2013 – February 2014) :
    a. All soil collected was sown in flats on top of 2” of standard potting soil, and exposed to early summer conditions (25/15C diurnal temps, watered as needed) in an incubator at Chicago Botanic Garden. Our aim was to determine whether any of the species in our study plots may already be present in the seed bank. Results are summarized in Appendix 1.
5. Vegetation monitoring (May and September 2013, June 2014, May 2016):
    a. We collected data on species presence and percent cover in all study plots three times over four years post-seeding: 1) May 2013, 2) June 2014, and 3) May 2016 following the same protocol as outlined in 1a. We do not anticipate any further data collection. In May 2016 all components of the study plot except for small wooden stakes in the ground were removed. Data has been entered into a database and analyzed to determine whether seeding treatments impacted the percent cover of native vegetation and non-native (*Bromus tectorum)* vegetation.
    b. We did not find any effects of seed mix treatment on the native plant cover or diversity of study plots, likely due in part to an extensive *Bromus tectorum* (cheatgrass) seedbank that was present in two of our three study blocks following the fire. We documented a rapid expansion of cheatgrass cover (up to 75%) over the 4 years of our study. At the block where cheatgrass wasn’t present, more than an inch of topsoil had been removed, leaving only a minimal seedbank. This site was on the rim of Horeshoe Canyon, and the wind seemed to have swept all seed away, including our applied treatments, as the site was nearly completely lacking in vegetation of any kind the year following the fire (we found *Helianthus annuus* flowering just down the hill from our study block, likely from the seeds we applied that were carried downhill by wind and water erosion).
    c. Wind & water erosion was also likely a confounding factor in our other two study blocks, as we identified some seeded species that weren’t aerially applied in our control plots during our 2013 vegetation survey.

The only species that was only found where it was applied was *Machaeranthera canescens*, which had seeds that were larger than many of our other species and it had been cleaned to remove the pappus, minimizing the potential to be dispersed via wind. Because of this, we report results in Appendix 1 by summarizing the number of plots that each species was identified in during our 3 vegetation surveys, as well as whether it was likely to be present in the soil seed bank and/or have seed rain from small pockets of remnant vegetation near study plots.

d. Of the species sown, all were present in at least one study plot over the course of our study except for the two Penstemon species (*Penstemon comarrhenus*, *Penstemon strictus* although one small *Penstemon* seedling was found in 2016 but not able to be identified because it was not flowering), *Elymus trachycaulis, Eriogonum umbellatum, Atriplex canescens, Sphaeralcea parvifolia,* and *Phacelia crenulata*.
e. Individual species and seed source responses are detailed in Appendix 1. Of note:
    i. Indian ricegrass (*Achnatherum hymenoides)* did not appear until 2 years post-seeding, likely due to seed dormancy identified in our seed germination trials. This is the only species that declined significantly between 2014 and 2016, and we have anecdotal evidence that the plants that died may have been Rimrock, while the wild-collected material from near Kanab, Utah continued to persist. However, this requires further investigation to be sure.
    ii. Hoary tansyasater (*Machaeranthera canescens)* and flax (*Linum spp.)* were the most successful forb species in our study if assessed by the number of plots where it was present during our 2016 survey. Because of seed drift and overseeding with the aerial application it is not clear which flax species was most often observed, but it is likely that most individuals recorded were blue flax, *Linum perenne* Appar.

Appendix 1: Species and sources used in each treatment, including seed germination trials. The number of plots that each species was found in over the four year monitoring time is included, along with notes for clarification. Results of seed germination trials are available via links above.

	Scientific name	Common name	Cultivar	Certification	Source	Seed Origin	Present 2012?	# Plots 2013	# Plots 2014	# Plots 2016
Part of "broad seed mix" applied aerially by GJFO	*Achillea millefolium var. occidentalis*	yarrow		SI	Rainier Seeds Inc		Seed Bank (Unburned)	0	16	22
	Achnatherum hymenoides	indian ricegrass	Rimrock	A	Maple Leaf			0	25*	16**
	Achnatherum hymenoides	indian ricegrass	Rimrock	A	North Basin Seed Company			0	25*	16**
	Achnatherum hymenoides	indian ricegrass	Rimrock	A	Rainier Seeds Inc			0	25*	16**
	Atriplex canescens	fourwing saltbush		SI	BLM Boise Warehouse	Kane, UT		0	0	0
	Elymus elymoides ssp. *elymoides*	bottlebrush squirreltail		SI	Native Seed Company	Duchesne, UT		2*	0	0
	Elymus trachycaulus	slender wheatgrass	San Luis	A	Landmark Turf & Native Seed			0	0	0
	Erigeron speciosus	aspen fleabane		SI	Uncompahgre Partnership	Colorado		0	2	2
	Eriogonum umbellatum	sulpher-flower buckwheat		SI	Uncompahgre Partnership	Colorado		0	0	0
	Linum lewisii	Lewis flax	Maple Grove	A	Western Reclamation			7***	17***	23***
	Linum perenne	blue flax	Appar	A	Rainier Seeds Inc			7***	17***	23***
	Pascopyrum smithii	western wheatgrass	Arriba	A	Central Milling Wheatland			0	0	34*
	Pascopyrum smithii	western wheatgrass	Arriba	A	Granite Seed Company			0	0	34*
	Pascopyrum smithii	western wheatgrass	Arriba	A	Landmark Turf & Native Seed			0	0	34*
	Penstemon comarrhenus	dusty beardtongue		SI	Uncompahgre Partnership	Colorado		0	0	0
	Poa secunda ssp. *secunda*	Sandberg bluegrass		SI	Granite Seed Company	Lincoln, WA		4*	0	0
	Sporobolus cryptandrus	sand dropseed		SI	Rainier Seeds Inc.	Rogersburg	In 3 plots	1*	22*	39*
	Sporobolus cryptandrus	sand dropseed		SI	Rainier Seeds Inc.	Snake River	In 3 plots	1*	22*	39*

	Scientific name	Common name	Cultivar	Certification	Source	Seed Origin	Present 2012?	# Plots 2013	# Plots 2014	# Plots 2016
Potential "native winners" applied in study plots	*Achnatherum hymenoides*	indian ricegrass		wild-collected	Seeds of Success (CBG2255)	Kane, UT		0	25*	16**
	Cleome lutea	yellow spiderflower		wild-collected	Seeds of Success UT931-453	Uintah, Utah		2	0	0
	Descurainia pinnata	western tansymustard		wild-collected	Seeds of Success CBG2272	Garfield, CO	Seed Bank (Burned)	5	33	0
	Elymus elymoides	bottlebrush squirreltail		wild-collected	Montrose BLM (K. Holsinger)	Montrose, CO		2*	0	0
	Elymus elymoides ssp. *elymoides*	bottlebrush squirreltail		wild-collected	Seeds of Success (CBG2252)	Mesa, UT		2*	0	0
	Helianthus annuus	annual sunflower		wild-collected	Seeds of Success (CBG2316)	Garfield, CO		3	5	0
	Heterotheca villosa	hairy false goldenaster		wild-collected	Seeds of Success (CBG2269)	Garfield, CO	Near plots (resprout post-fire)	0	0	8
	Machaeranthera canescens	hoary tansyaster		wild-collected	Seeds of Success (NM930N-62)	San Juan, New Mexico		0	8	22
	Mentzelia albicaulis	whitestem blazingstar		wild-collected	Seeds of Success (UT931-186)	Emery, Utah	Seed Bank (Unburned)	0	3	0
	Phacelia crenulata	cleftleaf wildheliotrope		wild-collected	Seeds of Success (UT931-449)	Emery, Utah		0	0	0
	Plantago patagonica	woolly plantain		wild-collected	Seeds of Success (UT93108-11)	Utah	Seed Bank (Unburned)	2	37	32
	Poa secunda ssp. *secunda*	Sandberg bluegrass		SI	Uncompahgre Partnership	Colorado		4*	0	0
	Sphaeralcea coccinea	scarlet globemallow		wild-collected	Seeds of Success (UT931-218)	Uintah, Utah	Near plots (resprout post-fire)	1	14	17
	Sphaeralcea parvifolia	small-leaf globemallows		wild-collected	Seeds of Success (UT080-50)	Uintah, Utah		0	0	0

*Multiple sources used and cannot determine success by source from vegetation data because this species was aerially sown over all plots as part of the BLM Broad Seed Mix.
**The plots where there was a loss from 2014 to 2016 were all sown with only 'Rimrock'
***Likely all *Linum perenne*, but not possible to tell because of seed movement and not all in flower.